

UNIVERSITY OF ENGINEERING AND TECHNOLOGY, LAHORE



ASSIGNMENT:

“SOFTWARE ENGINEERING”

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APPLIED COMPUTING

SECTION (B)

SUBMITTED TO: Mr. Maaz

INSTITUTE OF DATA SCIENCE, UET LAHORE

Software Engineering Assignment 01:

“Analysis of a Banking System”

A. Purpose

The primary purpose of a Core Banking System (CBS) is to provide a centralized, real-time environment that processes back-end daily financial operations, manages customer accounts, handles transactions, and facilitates electronic fund transfers. It ensures data consistency, transactional integrity, regulatory compliance, and seamless operational connectivity across physical branches, ATMs, mobile applications, and web interfaces.

B. Users

- 1. Retail & Corporate Customers:** Direct users who access digital banking apps, web portals, and ATMs to transfer funds, view account balances, and apply for financial products.
- 2. Bank Tellers & Branch Operations Staff:** Internal users who perform over-the-counter services such as cash deposits, ledger entries, wire transfers, and identity verification.
- 3. System Administrators & IT Engineers:** Technical staff responsible for infrastructure maintenance, access control setup, database operations, and system uptime.

C. Stakeholders

- 1. Financial Regulators & Central Banks:** Authorities enforcing financial laws, anti-money laundering (AML) directives, data protection, and operational reporting standards.
- 2. Bank Executive Leadership & Board:** Executives focused on risk management, operational costs, profitability, system resilience, and business growth.
- 3. Third-Party Payment Gateways & Partners:** External clearing houses, card networks (e.g., Visa, Mastercard), and fintech providers integrated via APIs.

D. Functional Requirements

- 1. Account Lifecycle Management:** The system shall allow creation, modification, freezing, and closing of customer savings, checking, and loan accounts.
- 2. Fund Transfer & Transaction Processing:** The system shall process real-time intra-bank and inter-bank fund transfers with atomic ACID (Atomicity, Consistency, Isolation, Durability) compliance.
- 3. Loan & Interest Calculation Engine:** The system shall automatically compute interest accruals, monthly installment schedules, and late payment penalties across various loan products.

4. KYC & Identity Verification: The system shall collect, store, and cross-reference customer identity records (e.g., CNIC/SSN, biometric data) during account onboarding.

5. Audit Logging & Transaction History: The system shall log every database mutation with timestamp, IP, and user ID to generate immutable audit trails for compliance.

E. Non-Functional Requirements

1. High Availability: The system must maintain 99.999% uptime (maximum ~5 minutes of unplanned downtime per year).

2. Low Latency: End-to-end API response time for transactional requests must be under 200 milliseconds under normal loads.

3. Data Encryption: All sensitive customer and financial data must be encrypted using AES-256 at rest and TLS 1.3 in transit.

4. Scalability: The architecture must horizontally scale to handle peak transaction loads (e.g., payday bursts) up to 10,000 transactions per second (TPS).

5. Regulatory Compliance: The system must comply strictly with international standards like PCI-DSS, GDPR, and local central bank regulations.

F. Risks

1. Cybersecurity Attacks & Data Breaches: Malicious actors targeting database endpoints or APIs could compromise customer identity and cause financial loss.

2. Core Database Corruption / Outage: Unscheduled server failures or concurrent transaction locks could halt branch operations and lock access to accounts.

3. Data Migration Failure during Upgrades: Errors in converting or migrating legacy data to modern systems could result in lost balances or mismatched accounting ledgers.

G. Quality Attributes

1. Security: Protecting system assets from unauthorized access, ensuring integrity of ledgers, and preventing fraud.

2. Reliability: Consistently performing accurate financial transactions without errors under continuous 24/7 operation.

3. Fault Tolerance: Ability of the system to maintain partial functionality or gracefully fail over to secondary nodes when a sub-component experiences hardware or software failures.

H. Where AI Could Be Used

- 1. Real-Time Fraud Detection:** Machine learning algorithms analyzing spending patterns, geolocation shifts, and transaction velocity to flag or block fraudulent activity.
- 2. Credit Scoring & Automated Underwriting:** AI models processing non-traditional customer data and financial history to assess creditworthiness and accelerate loan approvals.
- 3. Personalized Customer Assistants:** Generative AI-driven conversational agents handling routine account queries, dispute initiation, and personalized financial planning advice.

I. What Could Go Wrong if AI is Used Incorrectly?

- 1. Algorithmic Bias & Discrimination:** Mis calibrated credit scoring models could systemically discriminate against specific demographic groups during loan evaluations, violating fair lending laws.
- 2. AI Hallucinations & Financial Misinformation:** Conversational AI interfaces might generate incorrect account details, fabricated interest rates, or false fee schedules, resulting in customer panic or legal liability.
- 3. Black-Box Decisioning & Non-Explain ability:** Uninterpretable deep-learning models making credit denials or flagging account suspensions prevent the bank from satisfying regulatory mandates requiring transparent explanations to customers.